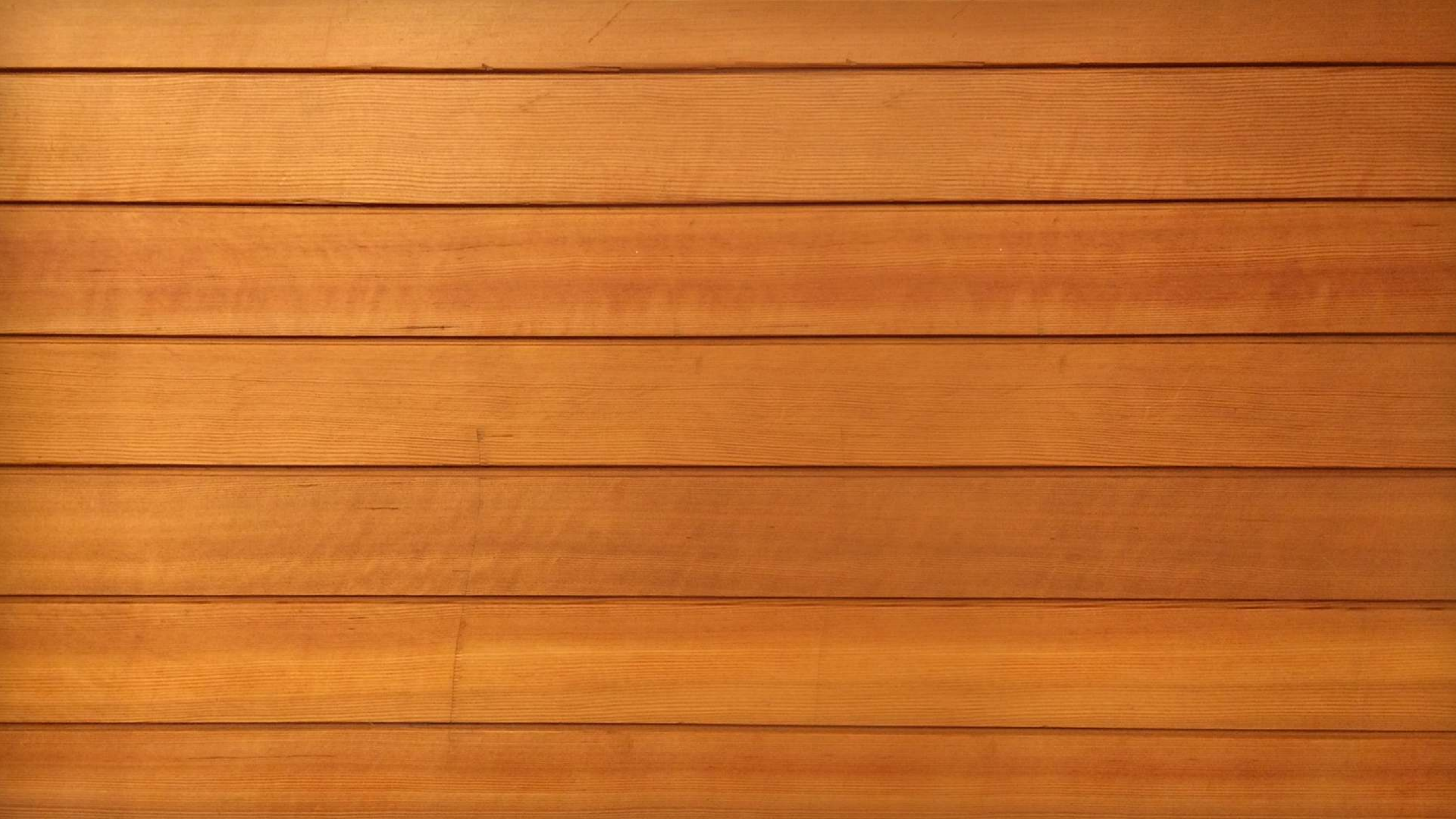






Est. 2011

*Grand
Cru*

Kallio, Helsinki

Who the ?

- Roope Kangas - Lead Server Developer @ Grand Cru
- Company founded in April 2011
- Our first game was Supernauts in 2014
- Four new projects after that - working on two new titles now



What I am going to talk about

- Before Elixir
- How we build our projects
- Learnings and our experience



Supernauts



Supernauts



Supernauts



```
Thread 2 (Thread 0x7ff8772ff700 (LWP 2406)):
#0 0x00007ff960255ab1 in sem_timedwait () from /lib64/libpthread.so.0
#1 0x000000000061fb7b in mono_sem_timedwait (sem=0x965fc8 <async_tp+40>, ti
#2 0x0000000000585632 in async_invoke_thread (data=0x0) at threadpool.c:156
#3 0x0000000000580de3 in start_wrapper_internal (data=0x7ff8c07d9000) at th
#4 start_wrapper (data=0x7ff8c07d9000) at threads.c:653
#5 0x0000000000614673 in thread_start_routine (args=0x7ff8cc0dd958) at wthr
#6 0x0000000000624340 in inner_start_thread (arg=0x7ff804de5d60) at mono-th
#7 0x00007ff96024ff18 in start_thread () from /lib64/libpthread.so.0
#8 0x00007ff95ff85e0d in clone () from /lib64/libc.so.6
```

```
Thread 1 (Thread 0x7ff960d7f780 (LWP 12092)):
#0 0x00007ff95fed72e2 in sigsuspend () from /lib64/libc.so.6
#1 0x00000000005c4714 in suspend_thread (info=0x1dbd040, context=0x7fff8850
#2 0x00000000005c486f in suspend_thread (context=0x7fff8850fb00, info=<opti
#3 suspend_handler (sig=<optimized out>, siginfo=<optimized out>, context=0
#4 <signal handler called>
#5 0x00007ff95ff7ca3d in poll () from /lib64/libc.so.6
#6 0x00007ff937dd4bf3 in wait_for_any (shutting_down=<optimized out>, timeo
out>, signals=<optimized out>) at signal.c:355
#7 Mono_Unix_UnixSignal_WaitAny (_signals=0x7ff957fe7dc8, count=3, timeout=
#8 0x000000000409aa723 in ?? ()
#9 0x00000000001e54630 in ?? ()
#10 0x00007ff960d7f700 in ?? ()
#11 0x0000000000000003 in ?? ()
#12 0x00000000041e3af76 in ?? ()
#13 0x00007ff957fe7dd8 in ?? ()
#14 0x00007ff957a12d78 in ?? ()
#15 0x00007fff88510310 in ?? ()
#16 0x0000000000000003 in ?? ()
#17 0x00007ff957fe7da8 in ?? ()
#18 0x00007ff957a127f0 in ?? ()
#19 0x00000000001e83ef0 in ?? ()
#20 0x0000000000000001 in ?? ()
#21 0x00007ff957fe7dc8 in ?? ()
#22 0x00007ff957fe7dc8 in ?? ()
#23 0x00007fff88510800 in ?? ()
#24 0x000000000409aa544 in ?? ()
#25 0x00000000001deea10 in ?? ()
#26 0x00007fff88510800 in ?? ()
#27 0x0000000000000000 in ?? ()
```

=====
Got a SIGSEGV while executing native code. This usually indicates
a fatal error in the mono runtime or one of the native libraries
used by your application.
=====

Heap shot summary

```
Heap shot 0 at 181.646 secs: size: 163489048, object count: 100534, class count: 2575, roots: 30733
Bytes      Count  Average Class name
152974568   78  1961212 System.Byte[]
7 root references (4 pinning)
24 references from: System.Byte[][]
7 references from: System.IO.FileStream
5 references from: MongoDB.Bson.IO.BsonTrieNode<System.Int32>
4 references from: System.IO.StreamWriter
4 references from: System.Security.Cryptography.MD5CryptoServiceProvider
3 references from: System.IO.CStreamWriter
3 references from: Mono.Globalization.Unicode.SimpleCollator
2 references from: GrandCru.FastZLib.ZStream
2 references from: System.MonoCQItem
2 references from: log4net.Appender.TurkServerS3Appender
1 references from: LookupService
1 references from: System.IO.MemoryStream
1 references from: System.TermInfoReader
1 references from: GameConfig
1 references from: System.IO.CStreamReader
1 references from: MongoDB.Bson.IO.BsonTrieNode<System.Boolean>
1 references from: System.Reflection.Emit.ModuleBuilder
1 references from: MongoDB.Bson.IO.BsonChunk
1 references from: GrandCru.FastZLib.InfBlocks
1 references from: StaticPool.poolentry[]
3522368    28080    125 System.String
5091 root references (2 pinning)
24061 references from: System.String[]
14058 references from: GeneralTaskConfig
8015 references from: GrandCru.Util.StatAverages.Counter[]
6702 references from: Utils.StaticEntry5
5174 references from: System.Object[]
4470 references from: Cache.Entry<KTuple<System.String, System.String>, Utils.StaticEntry5>
3476 references from: Utils.StaticEntry6
2613 references from: Cache.Entry<KTuple<System.String, System.String, System.String>, Utils.StaticEntry6>
2510 references from: STuple<System.String, System.String, System.Type>[]
1000 references from: System.Reflection.MonoField
762 references from: ObjectSerializationHelper.SerializableFieldInfo
624 references from: ChatEntry
581 references from: MissionProgressionData
435 references from: Substance
400 references from: MongoDB.Bson.Serialization.BsonMemberMap
390 references from: Utils.StaticEntry1
384 references from: TurfObjectConfig
361 references from: Cache.Entry<System.String, Utils.StaticEntry12>
360 references from: Utils.StaticEntry12
357 references from: System.Reflection.MonoParameterInfo
278 references from: System.Collections.Hashtable.Slot[]
252 references from: System.Configuration.SectionInfo
200 references from: ShopConfigEntry
196 references from: Cache.Entry<KTuple<System.String, System.Int32>, Utils.StaticEntry1>
186 references from: System.MonoTypeInfo
167 references from: System.Text.StringBuilder
146 references from: System.Collections.Specialized.NameObjectCollectionBase._Item
129 references from: SquareUpgradeConfig
```

Size: 117 KB

Earlier

Later

Now



Supernauts



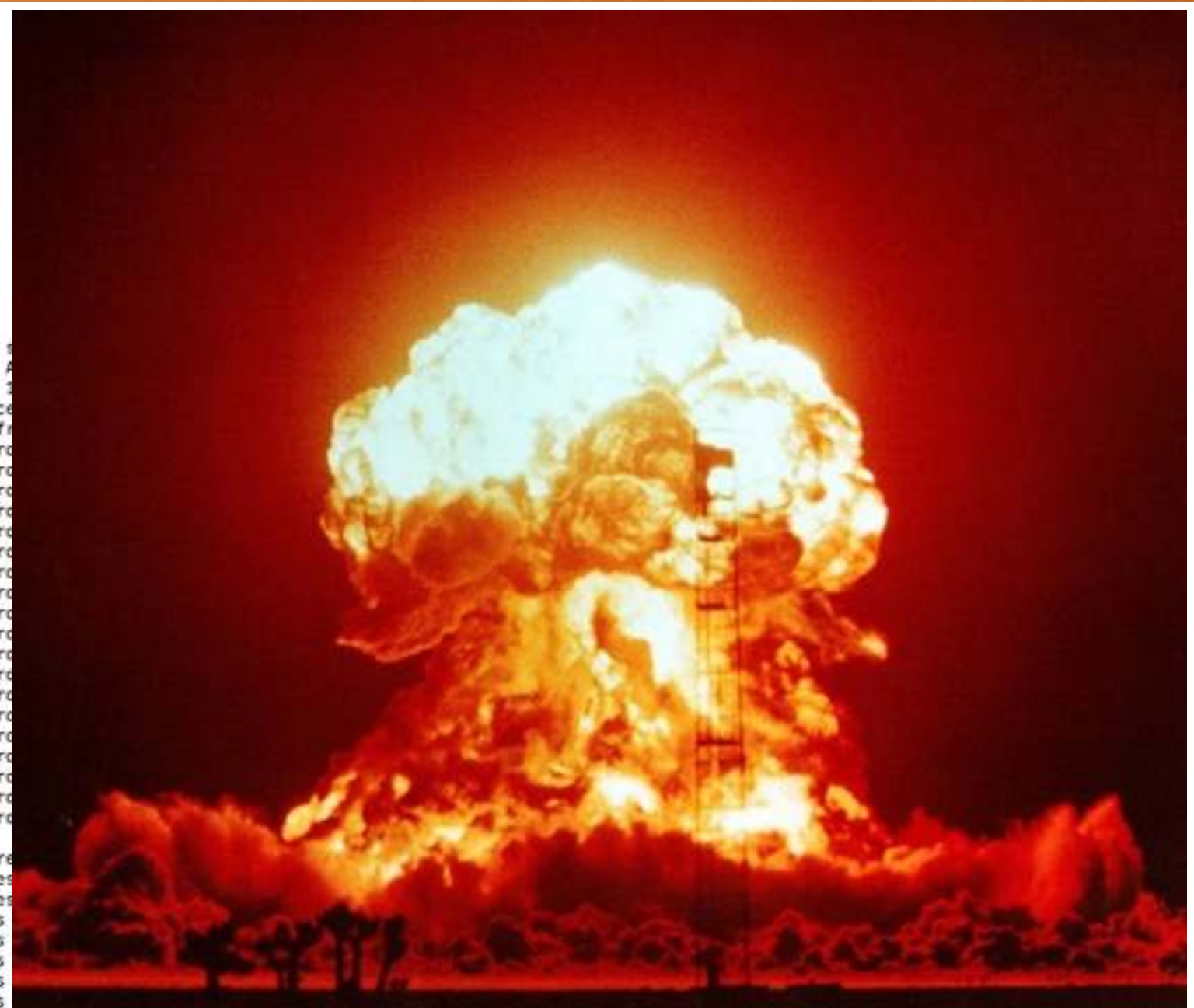
Thread 2 (Thread 0x7ff8772ff700 (LWP 2406)):
#0 0x00007ff960255ab1 in sem_timedwait () from /lib64/libpthread.so.0
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#3 0x0000000000580de3 in start_wrapper_internal (data=0x7ff8c07d9000) at th
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#7 Mono_Unix_UnixSignal_WaitAny (_signals=0x7ff957fe7dc8, count=3, timeout=!
#8 0x000000000409aa723 in ?? ()
#9 0x0000000001e54630 in ?? ()
#10 0x00007ff960d7f700 in ?? ()
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#15 0x00007fff88510310 in ?? ()
#16 0x0000000000000003 in ?? ()
#17 0x00007ff957fe7da8 in ?? ()
#18 0x00007ff957a127f0 in ?? ()
#19 0x0000000001e83ef0 in ?? ()
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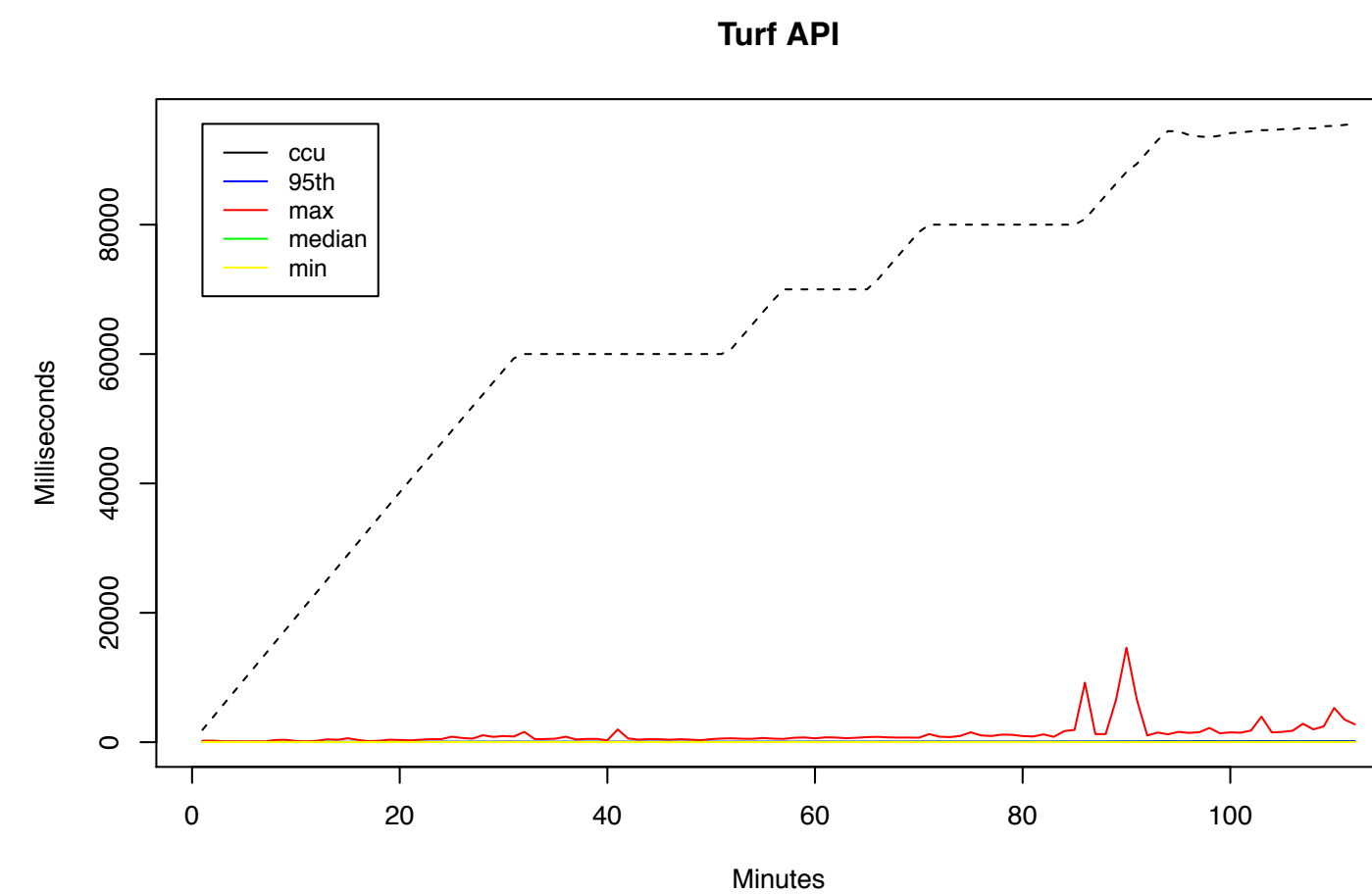
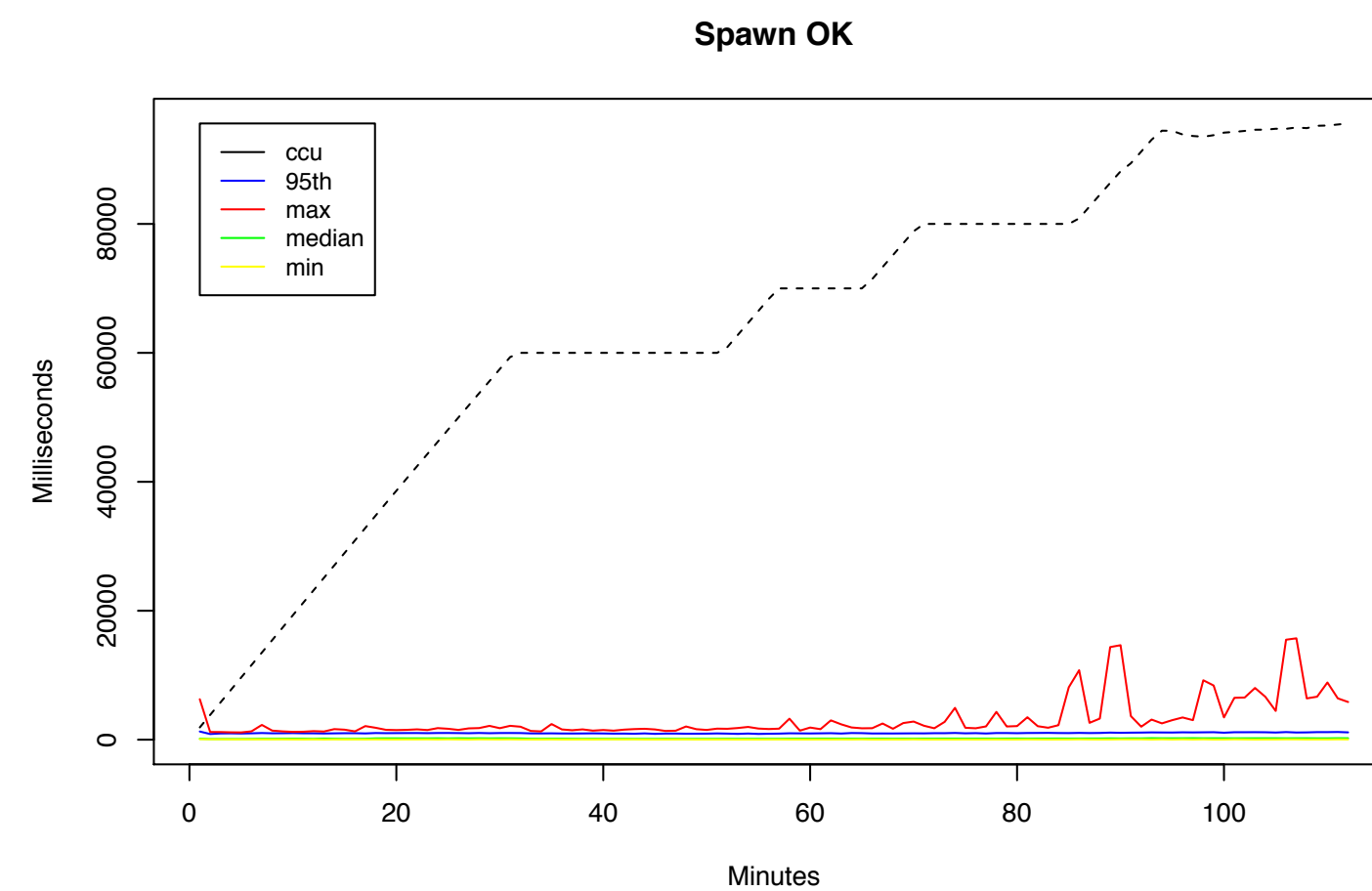
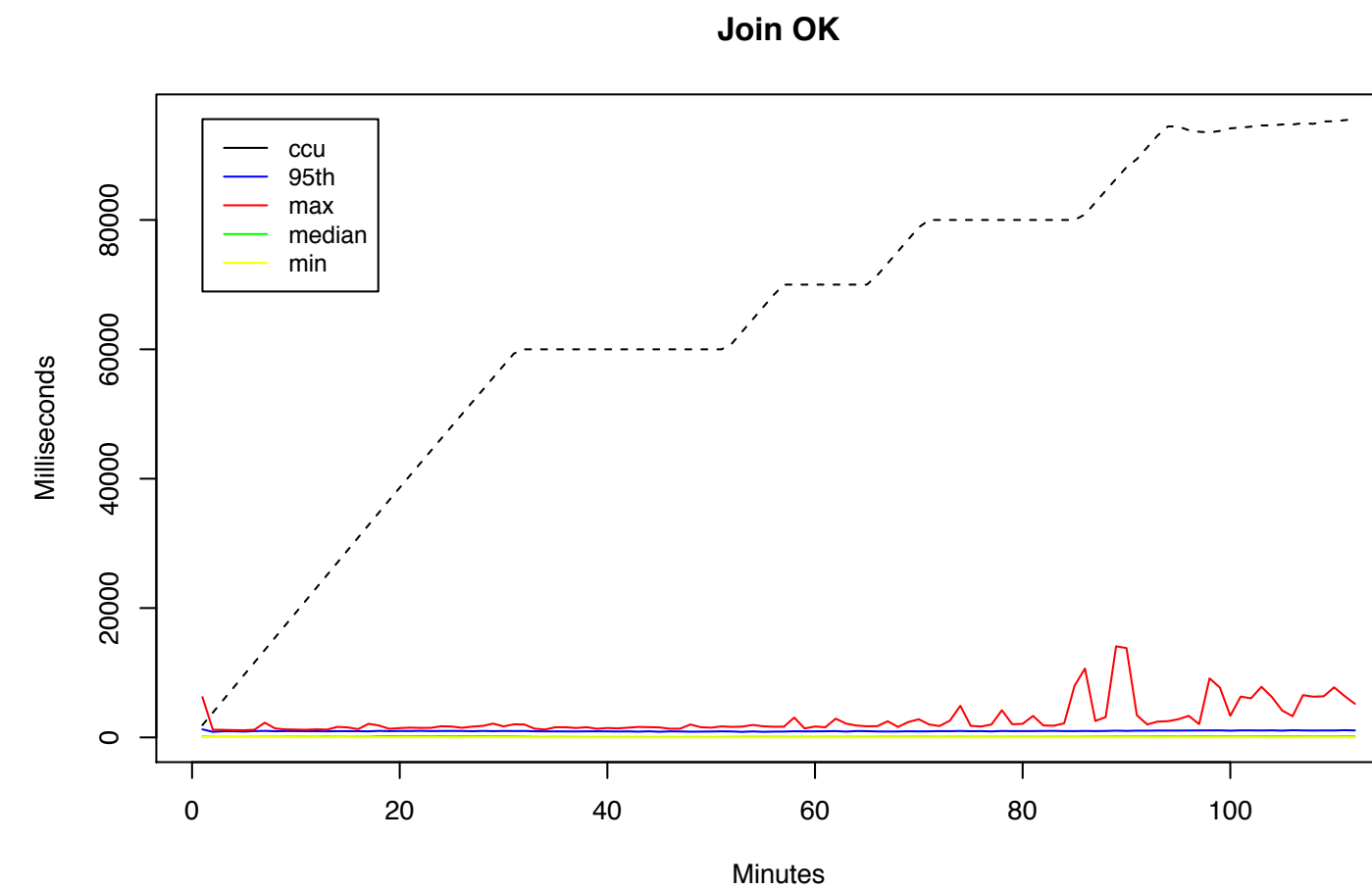
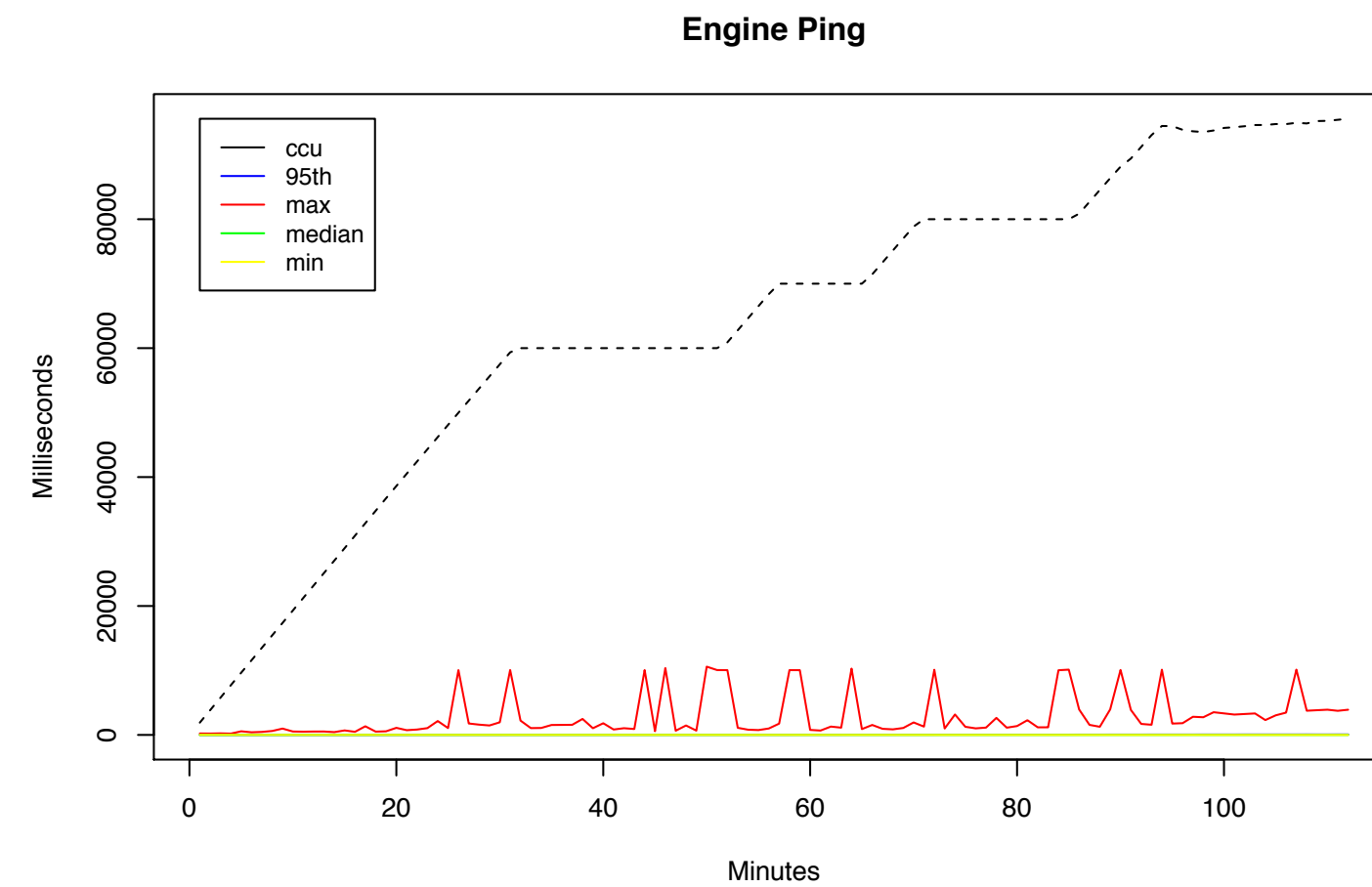
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Heap shot summary

Bytes	Count	Address
152974568	78	1
7	root reference	
24	references fr	
7	references fr	
5	references fr	
4	references fr	
4	references fr	
3	references fr	
3	references fr	
2	references fr	
2	references fr	
2	references fr	
1	references fr	
1	references fr	
1	references fr	
1	references fr	
1	references fr	
1	references fr	
1	references fr	
1	references fr	
1	references fr	
1	references fr	
3522368	28080	
5091	root refer	
24861	references	
14058	references	
8015	references	
6702	references	
5174	references	
4470	references	
3476	references	
2613	references	from: Cache.Entry<KTuple<System.String, System.String, System.String>, Utils.StaticEntry6>
2510	references	from: STuple<System.String, System.String, System.Type>[]
1000	references	from: System.Reflection.MonoField
762	references	from: ObjectSerializationHelper.SerializableFieldInfo
624	references	from: ChatEntry
581	references	from: MissionProgressionData
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400	references	from: MongoDB.Bson.Serialization.BsonMemberMap
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186	references	from: System.MonoTypeInfo
167	references	from: System.Text.StringBuilder
146	references	from: System.Collections.Specialized.NameObjectCollectionBase._Item
129	references	from: SquareUpgradeConfig



Used Erlang to load-test Supernauts



Learnings from Supernauts



Learnings from Supernauts

- Required a lot of "plumbing" to manage concurrency



Learnings from Supernauts

- Required a lot of "plumbing" to manage concurrency



Learnings from Supernauts

- Required a lot of "plumbing" to manage concurrency
- Chosen tech did not feel like a natural fit for the backend



Learnings from Supernauts

- Required a lot of "plumbing" to manage concurrency
- Chosen tech did not feel like a natural fit for the backend
 - Ended up doing lightweight processes with mailboxes by ourselves... :D

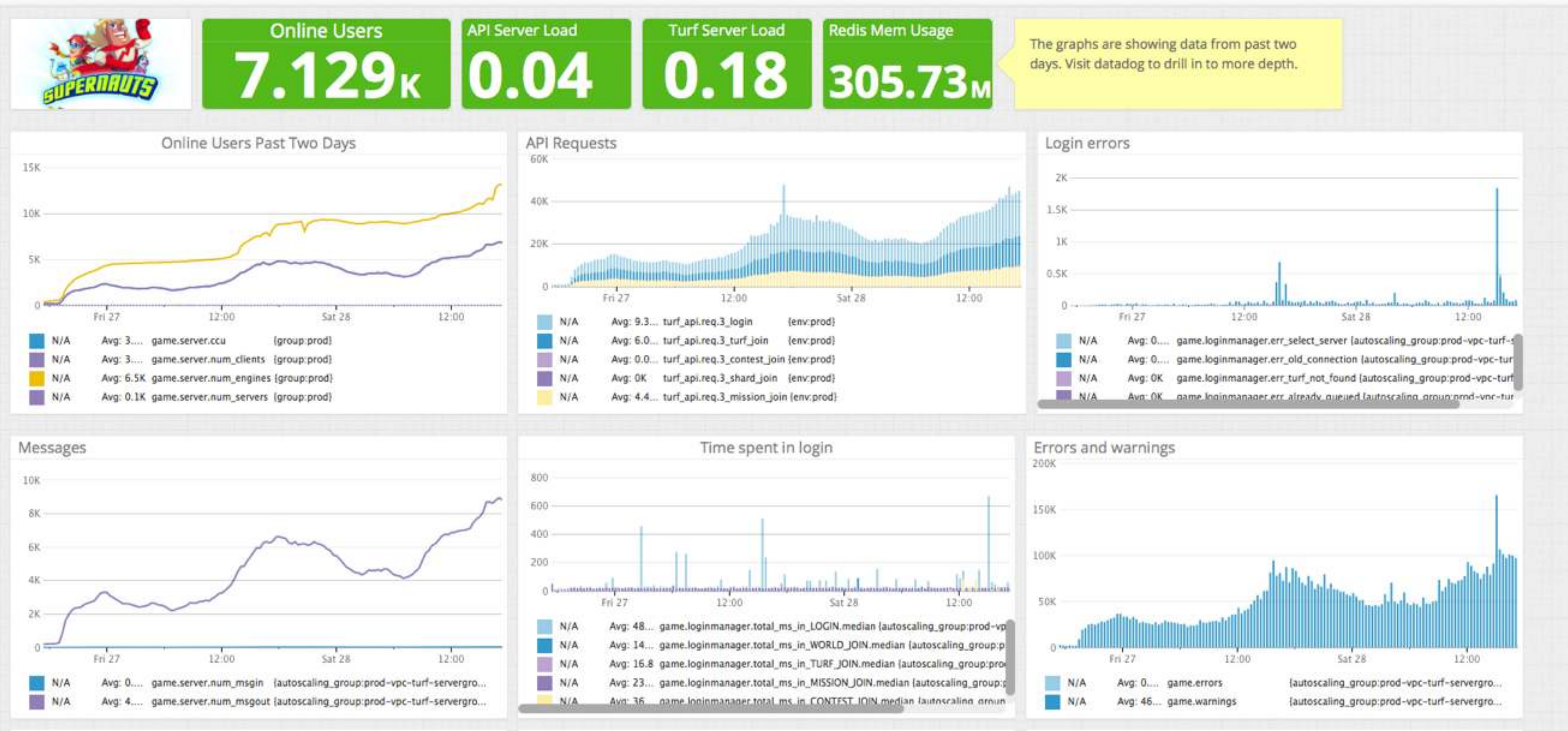


Supernauts



Supernauts

The Supernauts Dashboard



Grand Cru

Kallio, Helsinki

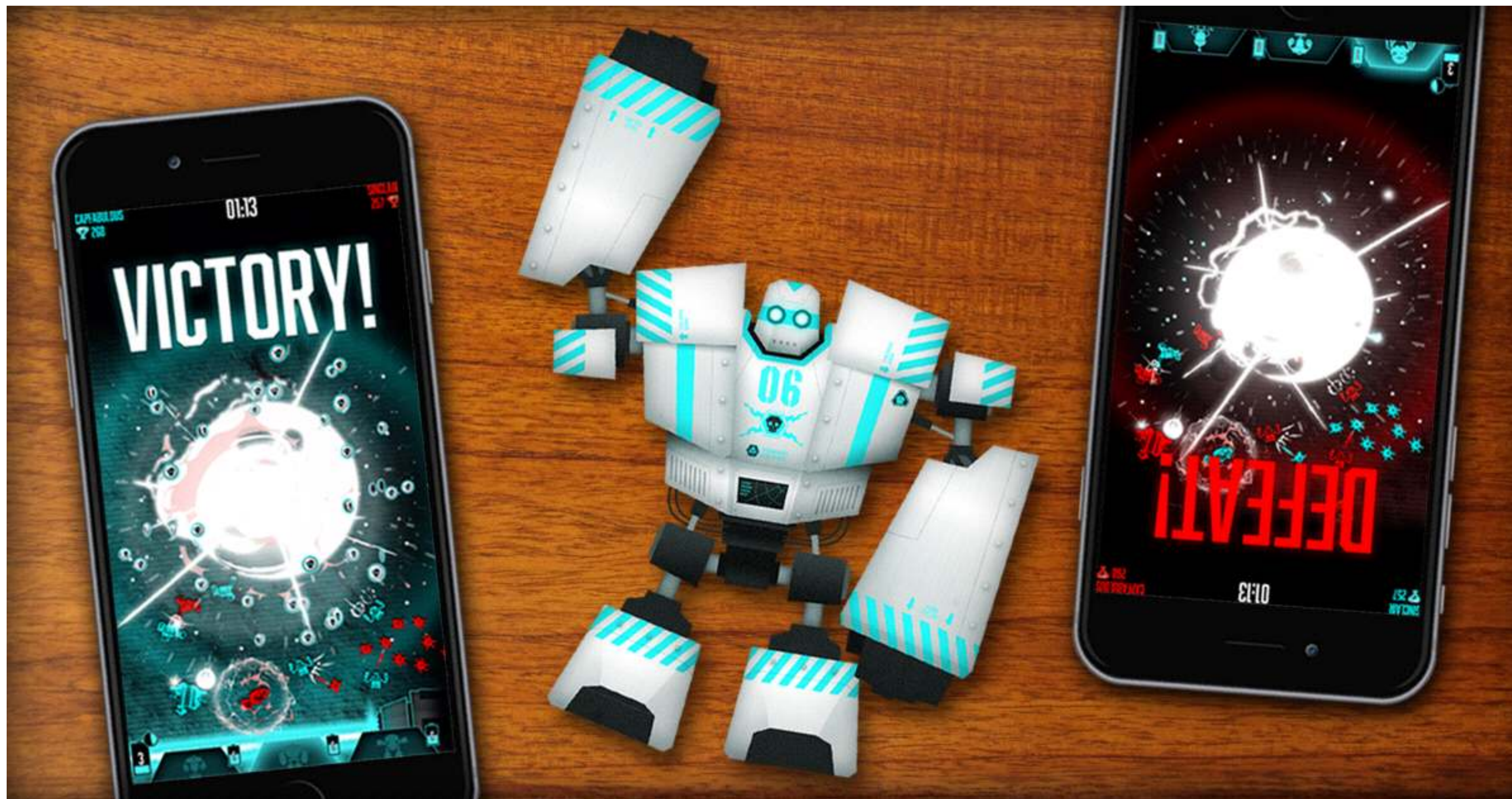
Our first sips



- 1) http://media.webcollage.net/rwvfp/wc/cp/19805765/module/tommeetippeeus/_cp/products/1420707514524/tab-6335b7db-4e6a-4b4a-89a4-c9bab7d2d19/dfbf5268-54fe-4457-bc91-8056bd7bde2b.jpg.w240.jpg
- 2) http://cdn.babygearlab.com/photos/0/58/306805_22066_XL.jpg



Warpfield 2014 <-> 2015



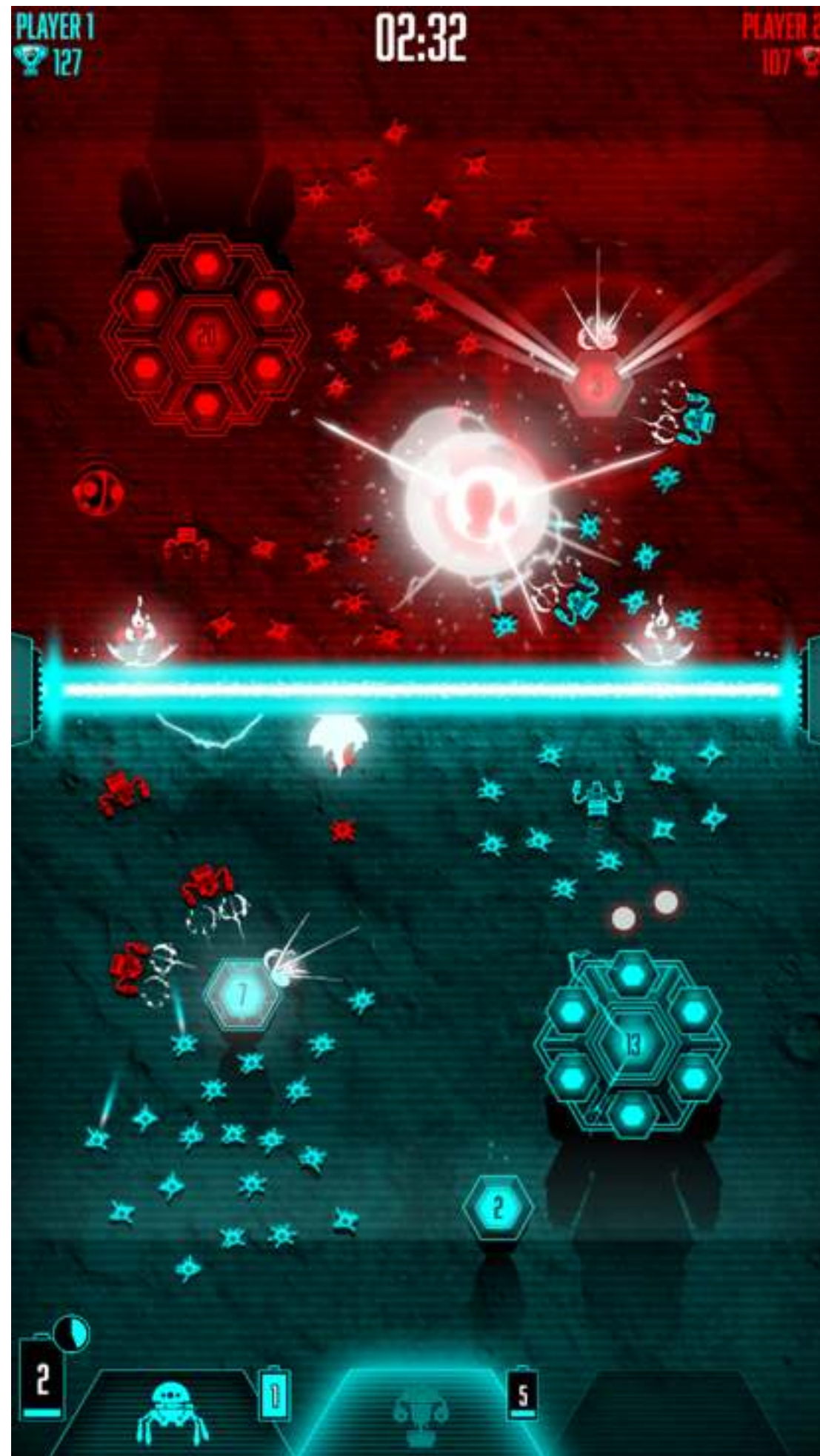
Warpfield - overview



- Player vs Player
- Crews and Tiered Leagues
- Free-to-play game
- Always connected



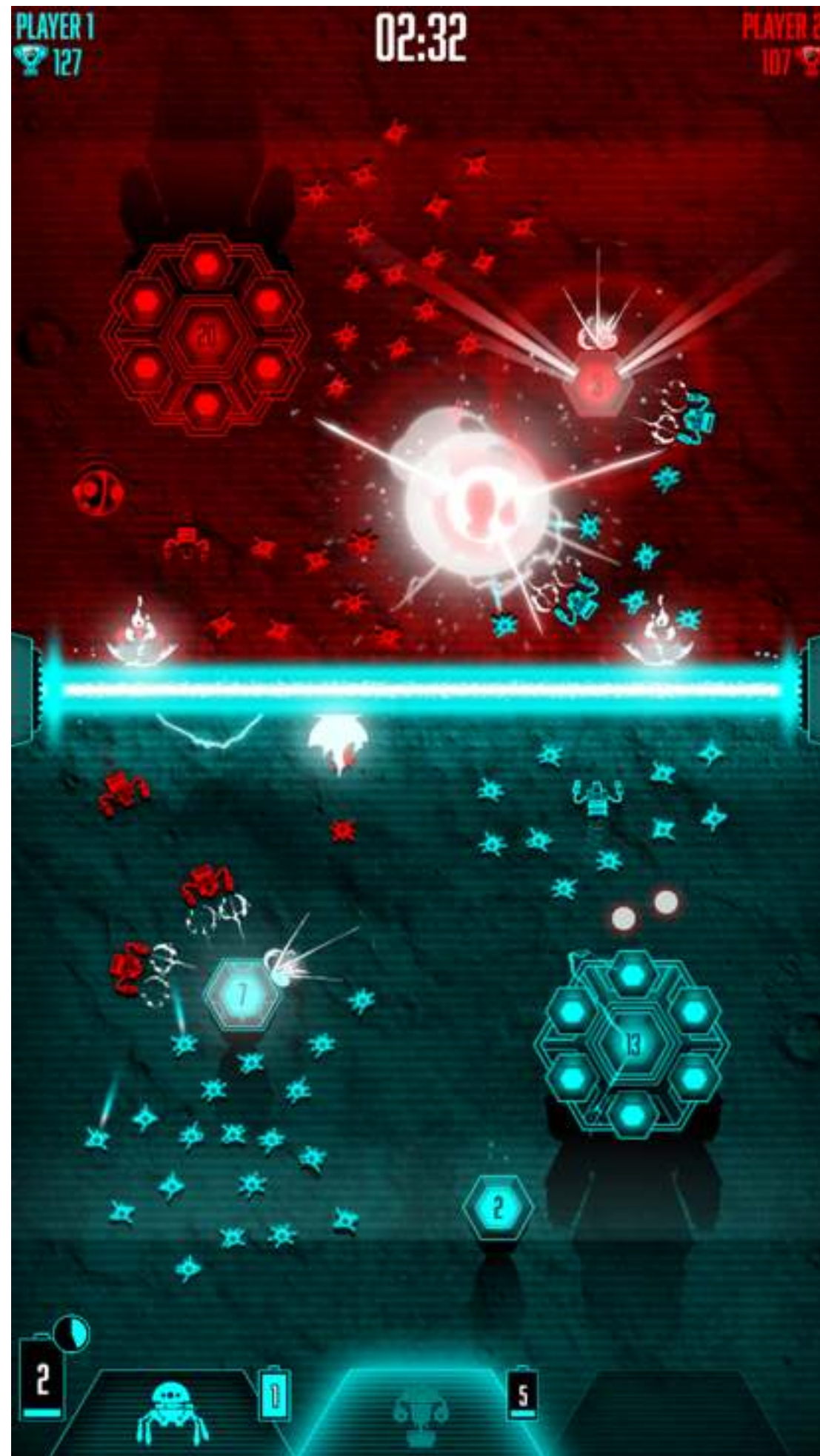
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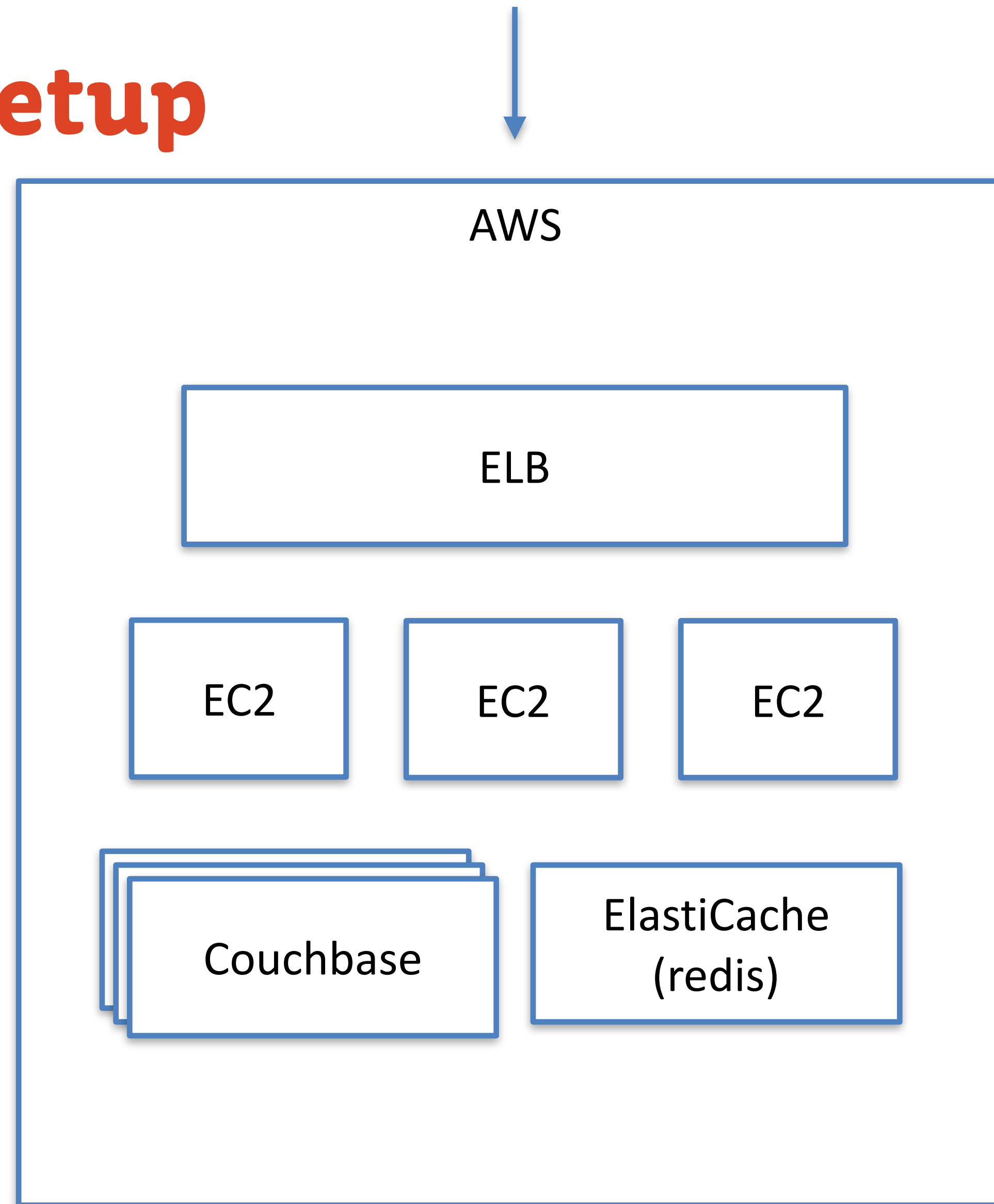
Warpfield - overview



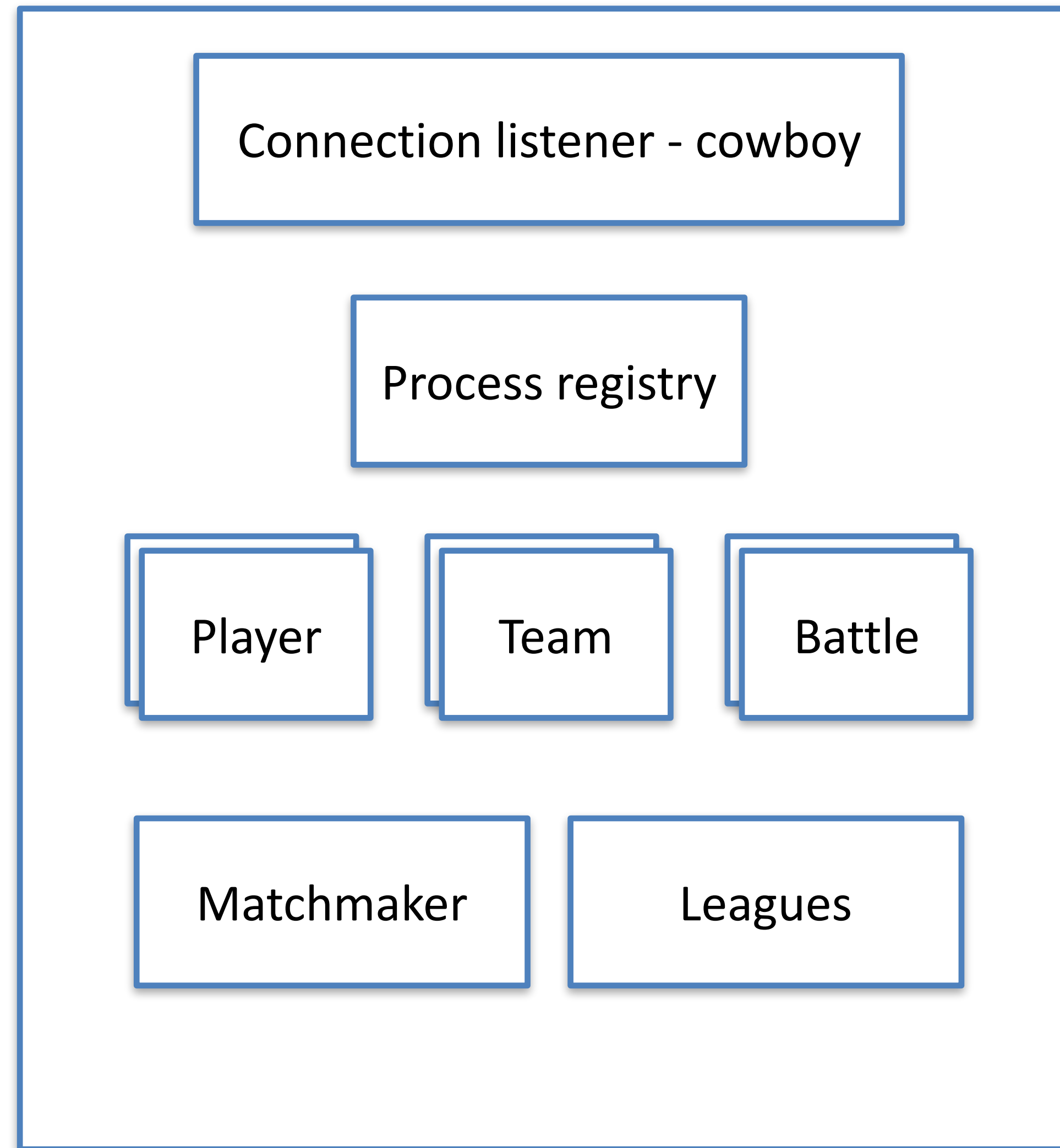
- Player vs Player
- Crews and Tiered Leagues
- Free-to-play game
- Always connected



Server setup



Inside the server



Finding the game process



Finding the game process

- GenServer supports custom registries



Finding the game process

- GenServer supports custom registries



Finding the game process

- GenServer supports custom registries

```
GenServer.start_link(__MODULE__, {:via, Relocker.Registry, name})
```



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Finding the game process

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GenServer.start_link(__MODULE__, {:via, Relocker.Registry, name})
```

```
# Node 1
```



Finding the game process

- GenServer supports custom registries

```
GenServer.start_link(__MODULE__, {:via, Relocker.Registry, name})
```

```
# Node 1  
{:ok, pid} = Game.start_link([], name: "player_1")
```



Finding the game process

- GenServer supports custom registries

```
GenServer.start_link(__MODULE__, {:via, Relocker.Registry, name})
```

```
# Node 1
```

```
{:ok, pid} = Game.start_link([], name: "player_1")
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```



Finding the game process

- GenServer supports custom registries

```
GenServer.start_link(__MODULE__, { :via, Relocker.Registry, name })
```

```
# Node 1  
{ :ok, pid } = Game.start_link([], name: "player_1")
```

```
# Node 2
```



Finding the game process

- GenServer supports custom registries

```
GenServer.start_link(__MODULE__, { :via, Relocker.Registry, name })
```

```
# Node 1  
{ :ok, pid } = Game.start_link([], name: "player_1")
```

```
# Node 2  
GenServer.cast({ :via, Relocker.Registry, "player_1" }, :stop)
```



League system

Season ends: 12d 4h
TROPHIES WILL BE RESET AFTER EVERY SEASON, HIGHER TIERS REWARD MORE COINS!

STARBURST 900+

SOLAR 300+

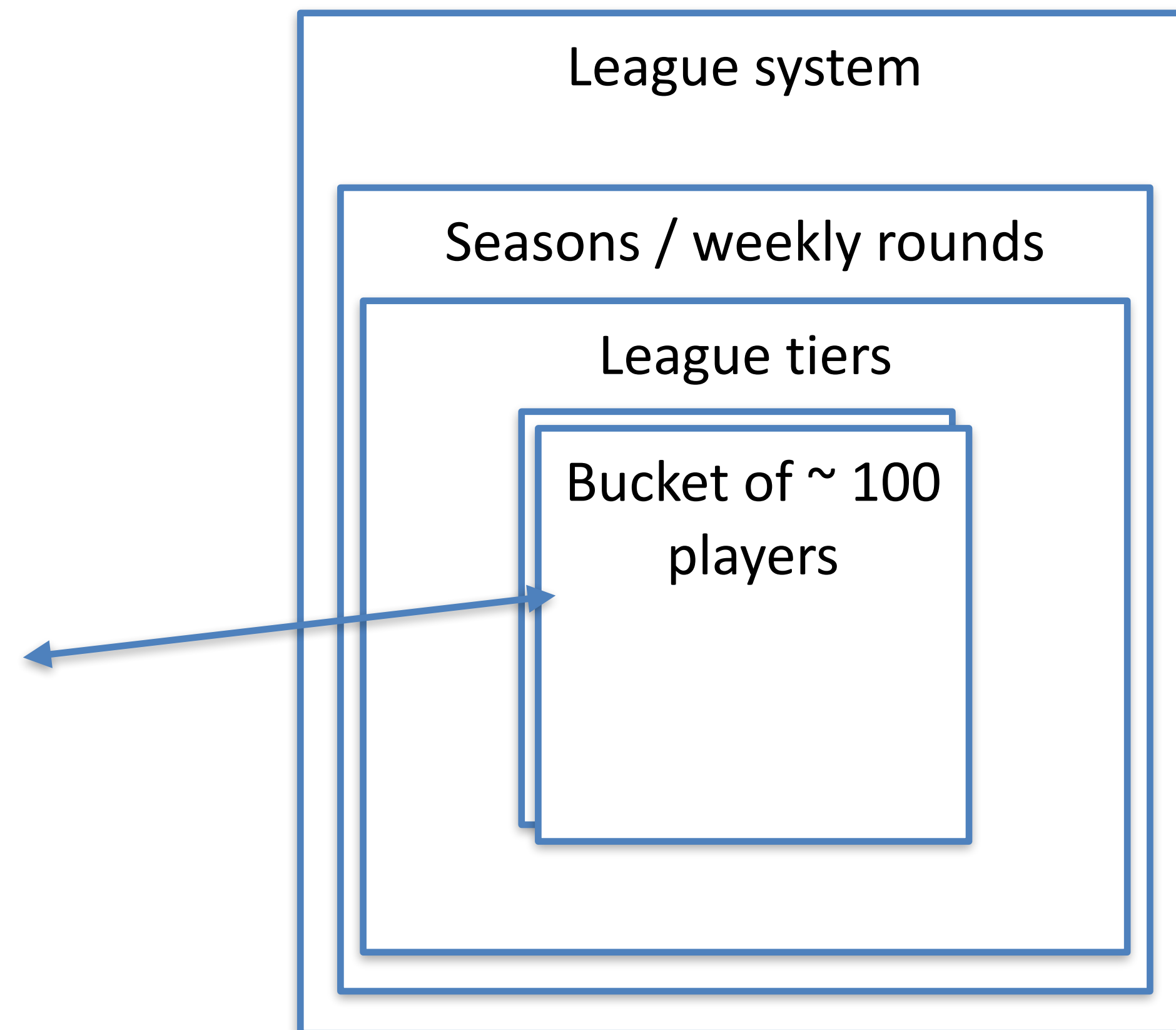
SOLAR I		700+
SEASON REWARD	6000	MATCH BONUS 165
1	[AI] ARRAY	884
2	[AI] ESMERALDA	819
3	[AI] MOLLY	787
4	??????	780
5	[AI] NATHALY	769
6	[AI] ZERO	754
7	[AI] MAJOR	739
8	[AI] BILL	715

SOLAR II 500+

SEASON REWARD 5000 MATCH BONUS 150

SOLAR III 300+

SEASON REWARD 4000 MATCH BONUS 125



Implementing Leagues with Elixir



Implementing Leagues with Elixir



Implementing Leagues with Elixir

- Built first without side-effects, sketch out the logic and tests



Implementing Leagues with Elixir

- Built first without side-effects, sketch out the logic and tests



Implementing Leagues with Elixir

- Built first without side-effects, sketch out the logic and tests
- For side-effects used in-memory implementation first with an Agent



Implementing Leagues with Elixir

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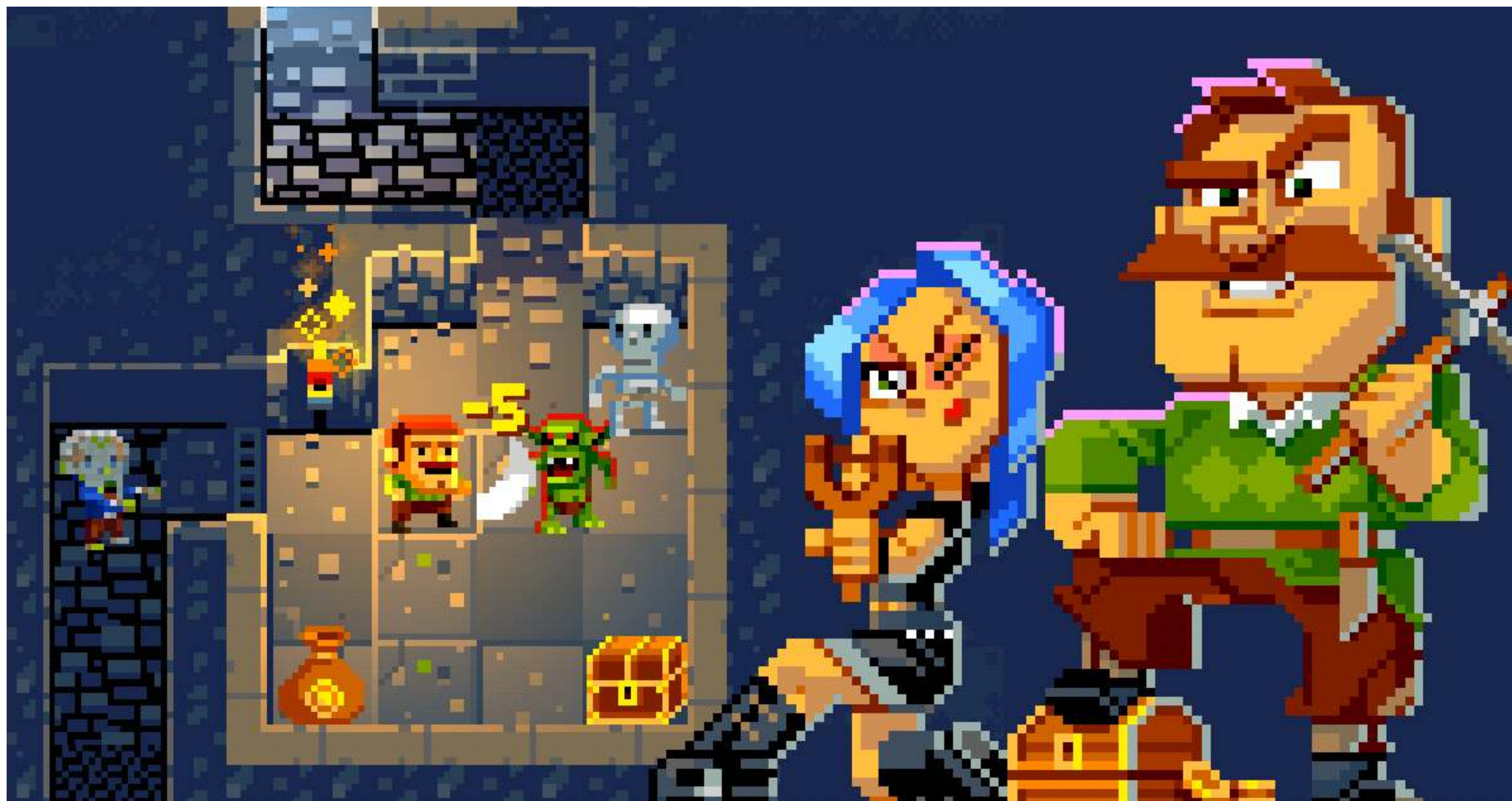


Implementing Leagues with Elixir

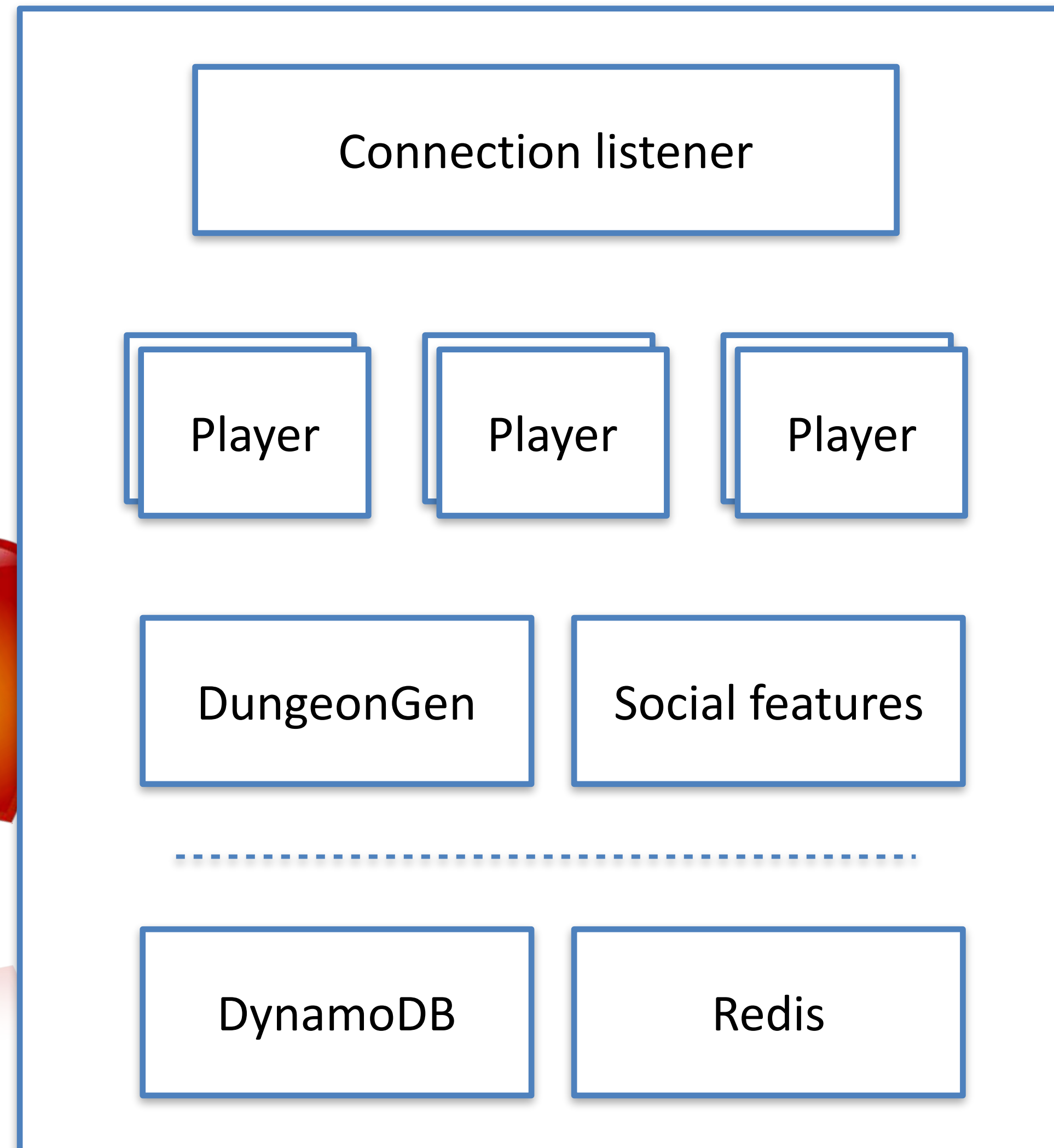
- Built first without side-effects, sketch out the logic and tests
- For side-effects used in-memory implementation first with an Agent
- Divide and conquer!



Hero Hack



Hero Hack Server



Hero Hack - client <-> server sync



Hero Hack - client <-> server sync

- Wire-protocol generated from Protobuf specification



Hero Hack - client <-> server sync

- Wire-protocol generated from Protobuf specification



Hero Hack - client <-> server sync

- Wire-protocol generated from Protobuf specification



Hero Hack - client <-> server sync

```
message C2S_Auth {  
    optional string device_model = 1;  
    optional string device_id = 2;  
    optional string config_hash = 3;  
    optional string device_name = 4;  
    optional bool fake_recap_events = 5;  
    optional string version = 6;  
    optional int64 client_time = 7;  
}
```



Hero Hack - client <-> server sync

```
var request = GameAPI.Instance.Auth(new C2S_Auth{...});  
yield return request.Start(); // wait for reply  
var response = request.Response;  
if (response.Ok) {  
    // \o/  
} else {  
    // :(  
}
```



Hero Hack - client <-> server sync

```
# Websocket process
def handle_message(%C2S_Auth{} = msg, req, %{state: :auth} = state) do
  player_id = PlayerRegistry.device_to_player!(msg.device_id)
  {:ok, pid} = GameSupervisor.find_or_start_game(args)
  GameProcess.handle_message(pid, msg)
end
```



Live config updates in dev



Live config updates in dev

- <https://github.com/GrandCru/GoogleSheets>



Live config updates in dev

- <https://github.com/GrandCru/GoogleSheets>



Live config updates in dev

- <https://github.com/GrandCru/GoogleSheets>
 - Poll config updates from Google Sheets to the game server



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Live config updates in dev

- <https://github.com/GrandCru/GoogleSheets>
 - Poll config updates from Google Sheets to the game server
 - Used in as part of our development process, production stuff separated



Elixir generated dungeons



Elixir generated dungeons

54	rule medium_rooms:		
55		wall_length	4..7
56			
57	rule large_rooms:		
58		wall_length	5..15
59			
60			
61	# DINGY DUNGEONS =====		
62			
63	area 1:		
64		medium_rooms	
65		width	20..25
66		height	20..25
67		proplevel	1
68		propamount	2
69		clutterlevel	1
70		hpower	4
71		door_chance	30
72		locked_door_chance	15
73		ammolevel	1
74			
75		dungeon 1:	
76		stages	2
77		mlevel	0
78		mpower	6..8
79		light	100
80		ammoprob	0%
81		locked_door_chance	0
82		chestlevel	1



Elixir generated dungeons

```
def generate_stage(conf, %GameConfig{} = gameconfig, rnd, opts \\ []) do
  # stage size
  width = number_or_random_range(rnd, conf.width)
  height = number_or_random_range(rnd, conf.height)

  # generate a random bsp and rooms from that
  bsp = BSPGen.generate_bsp(conf, rnd, {0, 0, width, height}, 6)
  rooms = generate_rooms(conf, rnd, gameconfig, bsp)

  # choose rooms which may contain traps and have the different floor type.
  # this is a subset of all rooms.
  rooms_with_traps = generate_rooms_with_traps(conf, rnd, gameconfig, rooms)

  # generate pathways to connect rooms
  pathways = generate_pathways(conf, rnd, rooms)
```



Elixir generated dungeons



Elixir generated dungeons



Load testing - setup



Load testing - setup

- Custom client application



Load testing - setup

- Custom client application
 - Simulates the game client, sends a fews messages - mostly idle



Load testing - setup

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 - Simulates the game client, sends a fews messages - mostly idle



Load testing - setup

- Custom client application
 - Simulates the game client, sends a fews messages - mostly idle
- Game servers **2** r3.xlarge 4 cores + 30GB RAM



Load testing - setup

- Custom client application
 - Simulates the game client, sends a few messages - mostly idle
- Game servers **2** r3.xlarge 4 cores + 30GB RAM
- DynamoDB 200/200 read/write ops for our devices & player tables



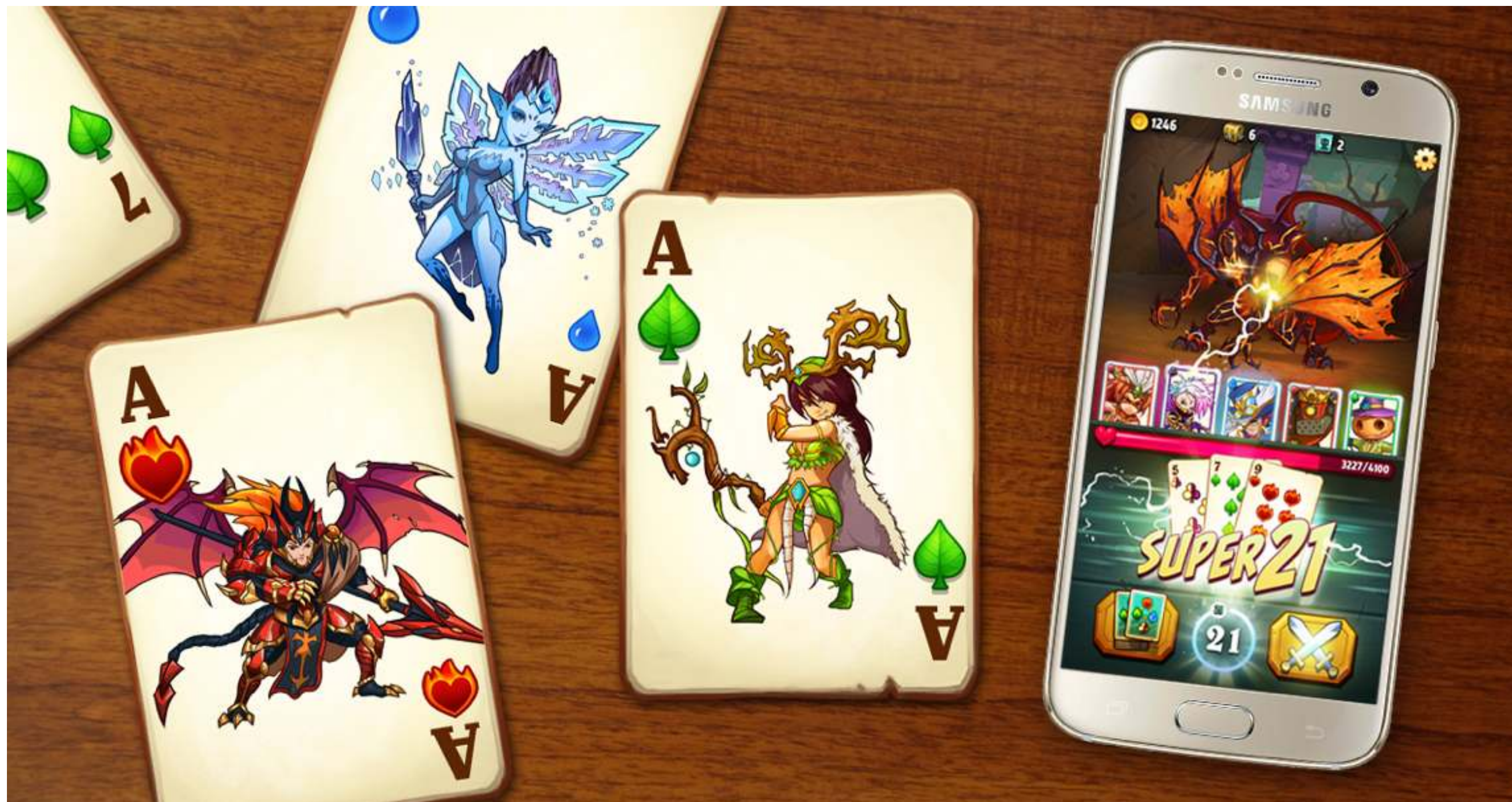
Load testing



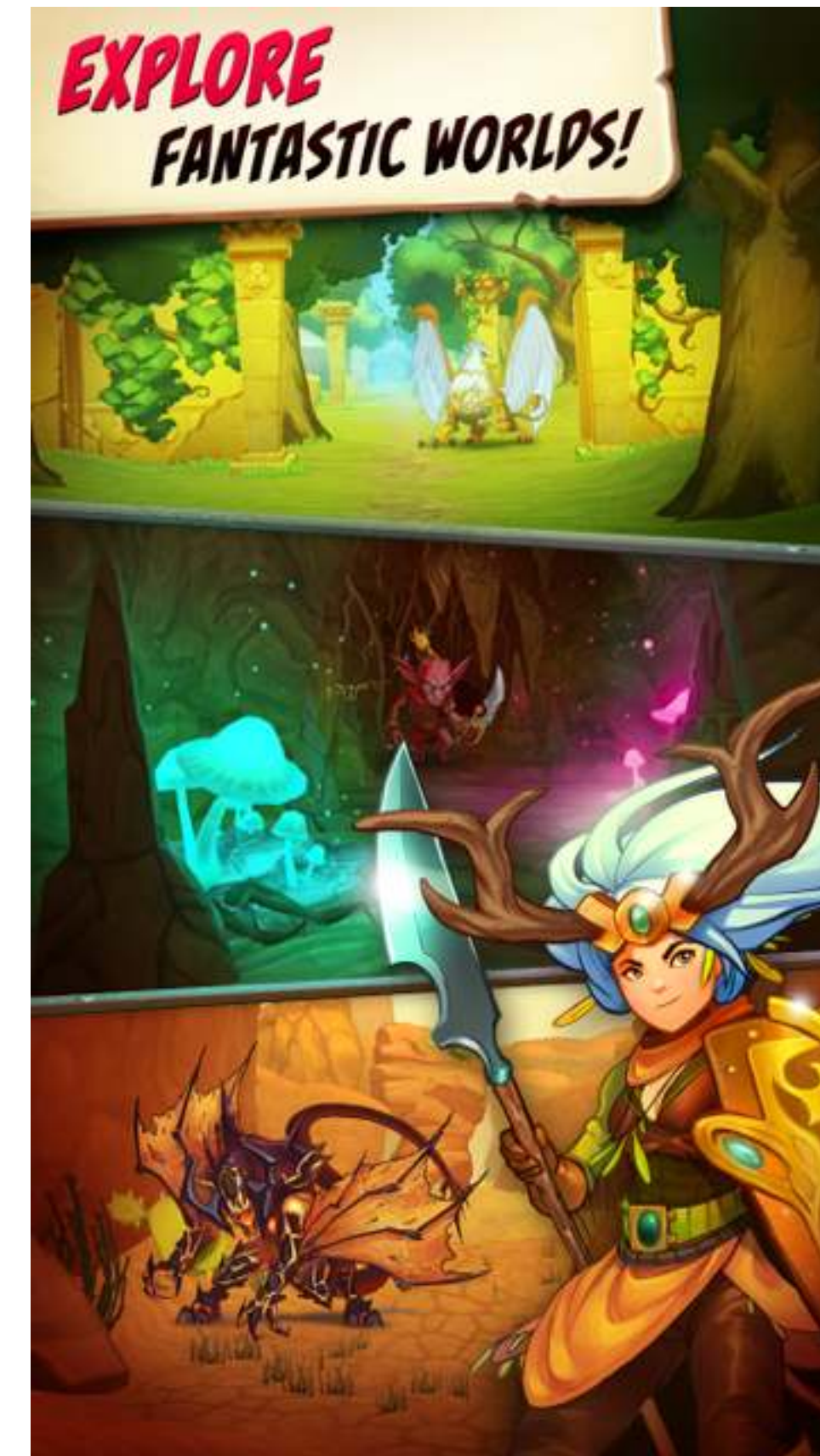
Load testing



Battlejack



Battlejack



From client driven to server drive



From client driven to server drive

- Client still predicts forward and assumes all operations succeed



From client driven to server drive

- Client still predicts forward and assumes all operations succeed



From client driven to server drive

- Client still predicts forward and assumes all operations succeed
- But while we are still working on the server, the action and resulting player state is sent to server



From client driven to server drive

- Client still predicts forward and assumes all operations succeed
- But while we are still working on the server, the action and resulting player state is sent to server

Client

```
%C2S_EvolveHero{hero_id:  
"hero", ..., expected_state:  
%PlayerState{}}
```



Server

```
Do evolve in Elixir, diff  
state -> if not  
expected_state  
report or explode
```



Server setup & tools



Server setup & tools

- Everything on AWS



Server setup & tools

- Everything on AWS
 - ELB, EC2, DynamoDB, Redis



Server setup & tools

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Server setup & tools

- Everything on AWS
 - ELB, EC2, DynamoDB, Redis
- Jenkins, builds releases



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Elixir in our experience



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- Elixir promotes code reuse - at least we have been able to share code between projects







Est. 2011

*Thank
You!*